Seungeun Lee, PhD

Founder & Principal · Dr. S Lee Life Sciences

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A scientist who codes and speaks the domain vocabulary. Fifteen years across research benches, regulatory bodies, and commercial leadership, with trusted life-science networks across Europe and Korea. Independent practice serving consulting firms, expert networks, and research organizations across EMEA and Asia.

Two current threads: **Senior Advisory** (verification and validation on advisory mandates, open across engagement levels and contract forms) and **Agentic AI** (Sense 2026, an agentic AI workspace by a life-sciences scientist, on active build).

01 EXPERIENCE

May 2026 – Present · Netherlands

Founder & Principal

Dr. S Lee Life Sciences

- **Commercial Strategy & Partnerships:** Europe–Asia go-to-market and partnership development for biotech. Market sizing, ICP definition, pricing and positioning, channel strategy, stakeholder mapping. Sales-operations data analysis. Multidisciplinary engagement leadership across sales, project management, lab, and bioinformatics.
- **Technical & Scientific Advisory:** R&D and research support across clinical, environmental, and food-safety life-sciences domains. NGS workflow design (in vivo, in vitro, in silico), method validation, proficiency testing, bioinformatics pipeline architecture (Python, R, Docker). Peer-reviewed manuscripts, technical reports, regulatory documentation.
- **Digital Infrastructure:** Database architecture and integration across technical and commercial operations data platforms. Reference database, biobank, LIMS, sales-operations, reporting infrastructure, ISO 17025 & ISO 9001 documentation and compliance. Platform-flexible: Agentic AI, Python, R, SAP, Power BI, SharePoint. Agentic AI workspace design (active build: Sense 2026).

Mar 2026 – May 2026 · Rotterdam

Entrepreneur in Residence

Builders

- Led domain research, ICP definition, and customer development for an early-stage AI SaaS. Structured interviews with scientific, commercial, and operations leaders across global R&D and biotech, surfacing buyer dynamics and market entry mechanisms.

02 EDUCATION

2016
PhD, Public Health
Seoul National University

2013
MSc, Molecular Medicine and Biopharmaceutical Science
Seoul National University

2011
BSc, Food and Nutrition · BSc, Biology
Yonsei University

03 SKILLS

- **Commercial strategy** — market entry, GTM, pricing, ICP, stakeholder mapping
- **Technical & scientific advisory** — NGS, bioinformatics, method validation
- **Digital infrastructure** — database architecture, LIMS, ISO 17025/9001
- **Agentic AI** — workspace design, LLM adoption
- **Cross-functional leadership** — multidisciplinary teams across R&D and commercial

04 CERTIFICATIONS

- PRINCE2 7 Practitioner
- ISO 17025 Awareness & Auditing
- IATA Dangerous Goods Regulation
- Big Data Insight
- Linux for Biologists

Rigor IN SCIENCE. *Clarity* IN STRATEGY.

05 LANGUAGES

- **Korean** — native

Feb 2021 – Feb 2026 · Amsterdam

Senior Project Manager · Head of Project Management

Macrogen Europe B.V.

- **Business development:** Commercialization of scientific innovations across EMEA. B2B/B2C partnerships, pricing and branding strategies for technical products.
- **Technical engagement:** Advised scientists on workflow trade-offs across in vitro and in silico methods. Led proficiency testing, validation reporting, data pipeline design.
- **Financial planning:** Fiscal planning, revenue forecasting, pricing optimization. Built reporting infrastructure with SAP, Power BI.
- **Digital infrastructure:** Designed organization-wide ISO 17025 and ISO 9001-compliant systems and LIMS workflows, improving traceability.
- **Leadership:** Managed multidisciplinary teams across sales, project management, lab operations, and bioinformatics.

Jan 2017 – Jan 2021 · Wageningen

Researcher Genomics · Data Scientist

Wageningen Food Safety Research (WFSR / NVWA)

- **Data analysis:** Genomic data analysis of food-borne pathogenic microbes using Illumina and Nanopore sequencing, with Python, R, Docker, Ubuntu, Git workflows.
- **Database management:** Developed biobank database infrastructure using LIMS, RDBMS (Microsoft Access), Jupyter Notebook, Python.

Apr 2016 – Oct 2016 · Bilthoven

Guest Researcher

PBL (Netherlands Environmental Assessment Agency)

- Curated metadata to refine the GLOBIO-Aquatic Projection model. Applied R packages, built regression models, aligned datasets through variable analysis.

Mar 2013 – Feb 2016 · Seoul

PhD Student Researcher

Environmental Health Engineering Lab, Seoul National University

- Built a new laboratory from the ground up to optimize in vitro and in silico workflows. Refined NGS biological sequence data with bioinformatics tools to identify environmental pathogens and improve microbial molecular identification.

Mar 2011 – Feb 2013 · Seoul

MSc Student Researcher

Genome Research Center for Diabetes and Endocrine Disease, SNU Hospital

- Molecular-based medical analyses: cell culture, differentiation, drug treatments, animal handling, qPCR, bioluminescent assays, immunoblotting, protein extraction, DNA cloning.

Jul 2009 – Dec 2009 · Davis, USA

Research Assistant

Western Human Nutrition Research Center, USDA, UC Davis

- Handled essential laboratory equipment, chemicals, and animal models for nutrition and health research applications.

- **English** — professional
- **Dutch** — conversational

06 TOOLS

- Python, R, Docker, SQL, Git
- SAP, Power BI, SharePoint
- LIMS, RDBMS, Jupyter
- Illumina, Nanopore (MinION)
- Agentic AI, LLM tooling

07 PUBLICATIONS AND REPORTS

Peer-reviewed publications

Indexed via ORCID with DOI

01 **Hoek A, Lee S, van den Berg R, Rapallini M, Overbeeke L, Opsteegh M, Bergval I, Wit B, van der Weijden C, van der Giessen J, van der Voort M.** (2023). Virulence and anti-microbial resistance of Shiga toxin-producing *Escherichia coli* from goat and sheep dairy farms in the Netherlands. *Journal of Applied Microbiology*.

02 **Lee S, An C, Xu S, Lee SH, Yamamoto N.** (2016). High-throughput sequencing reveals unprecedented diversities of *Aspergillus* species in outdoor air. *Letters in Applied Microbiology*. DOI: 10.1111/lam.12608.

Rigor IN SCIENCE. *Clarity* IN STRATEGY.

03 **Xu S, An C, Kim S, Lee S, Lee K, Yamamoto N.** (2016). Effects of the biocides on the culturable house dust-borne bacterial compositions and diversities. *Human and Ecological Risk Assessment*. 22(5): 1133-1146.

04 **Lee S, Yamamoto N.** (2015). Accuracy of the high-throughput amplicon sequencing to identify species within the genus *Aspergillus*. *Fungal Biology*. 119(12): 1311-1321.

05 **Lee S, Xu S, C.P. B, Lee H, Park MS, Lim YW, Yamamoto N.** (2015). Triazole susceptibilities in thermotolerant fungal isolates from outdoor air in the Seoul Capital Area in South Korea. *PLoS ONE*. 10(9): e0138725. DOI: 10.1371/journal.pone.0138725.

06 **Lee S, Kang S, C.P. B, Yoon C, Yang J, Yamamoto N.** (2015). Removal of viable airborne fungi from indoor environments by benzalkonium chloride-based aerosol disinfectants. *Human and Ecological Risk Assessment*. 21: 2174-2191.

07 **Lee SE, Koo YD, Lee JS, Kwak SH, Jung HS, Cho YM, Park YJ, Chung SS, Park KS.** (2015). Retinoid X receptor α overexpression alleviates mitochondrial dysfunction-induced insulin resistance through transcriptional regulation of insulin receptor substrate 1. *Molecules and Cells*. 38(4): 356-361.

Indexed at ORCID 0009-0009-3580-4459.

EU Reference Laboratory studies

Inter-laboratory proficiency studies

01 Lee S, van der Voort M. (2020). Global Microbial Identifier Proficiency Test 2017. *National Food Institute, Technical University of Denmark (DTU Food)*.

02 Lee S, van der Voort M. (2020). Evaluation on the first inter-laboratory exercise on Whole Genome Sequencing of Shiga toxin-producing *Escherichia coli* strains 2017-2018 (PT-WGS1). *European Union Reference Laboratory (EURL)*.

03 Lee S, van der Voort M. (2020). Evaluation on the 23rd inter-laboratory study (PT23) on the identification and typing of Shiga toxin-producing *E. coli* (STEC) and other pathogenic *E. coli* strains, 2018-2019. *European Union Reference Laboratory (EURL)*.

04 Lee S, van der Voort M. (2020). Evaluation on the 26th inter-laboratory study (PT26) on the identification and typing of Shiga toxin-producing *E. coli* (STEC), 2019. *European Union Reference Laboratory (EURL)*.

Internal technical reports

WFSR / NVWA · internal, not for external distribution

01 Lee S, van der Voort M. (2020). WGS Quality control: BaseClear reads reporting failure in the Per base sequence content. N200100. WFSR.

02 Lee S, Bak A, van der Voort M. (2020). Onderzoeksrapport: Optimization of Nanopore whole genome sequencing. Assembly assessment. R200086. WFSR.

03 Lee S, van der Voort M. (2020). Onderzoeksrapport: Implementing the trim workflow for the quality control of whole genome sequencing reads. R200057. WFSR.

04 Lee S, van der Voort M. (2020). Onderzoeksrapport: Implementing cross-contamination workflow for the quality control of *Escherichia coli* WGS data sets. R200056. WFSR.

05 Lee S, Castelijn G, van der Voort M. (2020). Onderzoeksrapport: Setting cut-off values for the quality control of *Listeria monocytogenes* WGS data sets. R190130. WFSR.

06 Lee S, van der Voort M. (2019). Onderzoeksplan: Optimization of the use of Nanopore whole genome sequencing. N190072. WFSR.

07 Lee S, Rapallini M, Bak A, van der Voort M. (2019). Onderzoeksrapport: Implementation of the Oxford Nanopore Sequencer MinION for Whole Genome Sequencing. R190021. NVWA.

08 Lee S. (2019). Biobank database update manual. NVWA.

09 Lee S, van den Berg R, van der Voort M. (2018). Onderzoeksrapport: Implementatie bioinformatica workflows voor (Shiga Toxin-producing) *Escherichia coli*. R180034. NVWA.

10 Lee S. (2017). LV8.T.Biobanking User Manual. NVWA.

Conference presentations

Selected talks · awards where noted

01 Lee S. Biobank training: working with scripts. Wageningen Food Safety Research, Wageningen, The Netherlands. Jul 30, 2020.

02 Lee S. Effects of global temperature change on freshwater fish: temperature sensitivity of freshwater fish. GLOBIO meeting, PBL Netherlands Environmental Assessment Agency. Aug 30, 2016.

03 Lee S, Yamamoto N. Transcriptome analysis of *Aspergillus fumigatus* exposed to triazole fungicides by RNA-Seq. 2nd BK Plus CHEER International Workshop, Seoul National University, Korea. Feb 14, 2016.

04 Lee S. Identifying the risks of triazole chemical uses on environmental fungal communities and properties. Graduate Research Competition, Graduate School of Public Health, Seoul National University, Korea. Mar 6, 2015. AWARDED

05 Lee S, Xu S, C.P. B, Lee H, Park MS, Lim YW, Yamamoto N. Detecting triazole-resistant fungi from ambient air in Seoul, South Korea. 1st BK Plus CHEER International Workshop, Seoul National University, Korea. Feb 2, 2015. AWARDED

06 Lee S, Kang S, C.P. B, Yoon C, Yang J, Yamamoto N. Removal Effects of Benzalkonium Chloride-based Aerosol Disinfectants on Indoor Airborne Fungi. International Aerosol Conference, Busan, Korea. Sep 2, 2014.

07 Lee S, Yamamoto N. NGS-based study of fungal allergy. Epidemiological analyses of Environment Seminar, Asan Medical Center, Seoul, Korea. Jan 8, 2013.

08 Lee S, Chung SS, Ahn BY, Koo YD, Park KS. Quantitative recovery of IRS1 with RXR α activation in mitochondrial dysfunction associated insulin resistance. International Conference on Diabetes and Metabolism. Nov 10, 2012. AWARDED

09 Lee S. Current Issue on Aerobiology and Allergy: An Environmental Approach to Public Health. Global Talent Program Workshop, UN Environmental Programme, Bangkok, Thailand. Jun 28, 2013.